

EventSphere - Low Level Design (LLD)

React + Spring Boot + Spring AI

1. Architecture

Presentation Layer (React) → REST/WebSocket → Spring Boot → Service Layer → Repository Layer → PostgreSQL.

Redis is used for caching. Spring AI handles AI features.

2. Backend Package Structure

com.eventsphere

- config
- security
- controller
- service
- repository
- entity
- dto
- mapper
- websocket
- ai
- notification
- exception
- util

3. Core Entities

User(id, name, email, password, role)

Organizer (inherits User role)

Event(id, title, description, location, date, capacity, organizerId)

Registration(id, userId, eventId, status)

Ticket(id, qrCode, registrationId)

Attendance(id, ticketId, checkInTime)

Connection(id, senderId, receiverId, status)

Message(id, senderId, receiverId, content, createdAt)

Notification(id, userId, type, message, readStatus)

4. Service Layer

AuthService

UserService

EventService

RegistrationService

TicketService

AttendanceService

ChatService

ConnectionService

NotificationService

AIService

5. Repository Layer

UserRepository
EventRepository
RegistrationRepository
TicketRepository
AttendanceRepository
ConnectionRepository
MessageRepository
NotificationRepository

6. DTOs

LoginRequest
RegisterRequest
EventRequest
EventResponse
RegistrationResponse
ChatMessageDTO
NotificationDTO

7. API Flow

Login → JWT Token
JWT attached in every request.
Organizer creates event → Event stored in PostgreSQL.
Participant registers → Registration + QR Ticket generated.
QR scanned → Attendance updated.
Chat uses WebSocket for live communication.
AI endpoints call Spring AI for description generation and recommendations.

8. React Folder Structure

src/
components/
pages/
layouts/
hooks/
context/
services/
api/
routes/
store/
types/
utils/

9. Security Design

Spring Security 6
JWT Authentication
BCrypt Password Encoding
Role Based Access:
ROLE_ADMIN
ROLE_ORGANIZER
ROLE_PARTICIPANT

10. Database Relationships

User 1:N Event (Organizer)
User M:N Event through Registration
Registration 1:1 Ticket
Ticket 1:1 Attendance
User M:N User through Connection
User 1:N Notification

11. Future Extensibility

Redis Cache
Cloudinary
Docker Compose
GitHub Actions
Kubernetes
Microservices (Auth, Event, Chat, AI)